

Artificial Intelligence Internship – Student Performance Prediction

Project Overview

This project was developed as part of the **CSE7000 – Internship on Artificial Intelligence** at Presidency University, Bengaluru.

The project, titled “**Student Performance Prediction Using Machine Learning**,” focuses on predicting student academic performance using factors such as attendance, study hours, assignment scores, previous examination marks, internal assessment scores, and classroom participation.

The project demonstrates the complete Machine Learning workflow, including data collection, preprocessing, model training, testing, evaluation, and prediction.

Objectives

- Collect and preprocess educational datasets.
- Analyze factors affecting student academic performance.
- Build Machine Learning models for prediction.
- Compare different Machine Learning algorithms.
- Evaluate model performance using standard metrics.
- Gain practical experience in Artificial Intelligence and Machine Learning.

Technologies Used

- **Programming Language:** Python
- **Machine Learning:** Scikit-learn
- **Data Processing:** Pandas, NumPy
- **Data Visualization:** Matplotlib
- **Deep Learning:** TensorFlow, Keras
- **Computer Vision:** OpenCV
- **Natural Language Processing:** NLTK

Dataset Features

The project uses student-related academic information, including:

- Student ID
- Attendance Percentage
- Study Hours
- Assignment Scores
- Previous Semester Marks
- Internal Assessment Scores



Project Workflow

1. Problem Identification
2. Data Collection
3. Data Preprocessing
4. Feature Engineering
5. Dataset Splitting
6. Model Training
7. Model Testing
8. Performance Evaluation
9. Prediction



Machine Learning Algorithms

The following algorithms were studied and/or implemented:

- Decision Tree
- Random Forest
- Logistic Regression
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Linear Regression
- Naïve Bayes
- K-Means Clustering



Data Preprocessing

The dataset was prepared through the following steps:

- Removing duplicate records
- Handling missing values
- Encoding categorical variables
- Normalizing numerical data
- Splitting data into training and testing datasets



Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics help determine the effectiveness of the trained models in predicting student performance.

Internship Learning Outcomes

The internship provided practical exposure to:

- Python programming
- Machine Learning
- Deep Learning
- Data preprocessing
- Feature engineering
- Data visualization
- Model training and evaluation
- Natural Language Processing
- Computer Vision
- AI project development

The internship also improved problem-solving, analytical thinking, programming, documentation, teamwork, communication, and time-management skills.

Organization

Alfido Tech Pvt. Ltd.

The organization focuses on Artificial Intelligence, Machine Learning, Data Science, Cloud Computing, and Software Development. The internship provided exposure to practical AI technologies, datasets, coding activities, and model development.

Student

Name: Monisha G

Roll Number: 20241CSG0013

Program: Bachelor of Technology in Computer Science and Engineering

University: Presidency University, Bengaluru

Conclusion

The Artificial Intelligence internship provided practical knowledge of AI and Machine Learning and helped bridge the gap between academic concepts and real-world applications. The Student Performance Prediction project demonstrated how Machine Learning can be used to analyze educational data and develop predictive solutions.

The knowledge gained through this internship provides a foundation for further learning and professional development in Artificial Intelligence, Machine Learning, Data Science, Software Engineering, and Intelligent Automation.